

1.7 Quiz Solutions: Measurement

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once.

1. Significant digits.

(a) **3**

(c) **2**

(e) **1**

(b) **3**

(d) **4**

(f) **3**

(a) Leading zeros do not count; the trailing zero after the decimal point does. (b) A zero between nonzero digits always counts. (e) 0.000 09 has a single measured digit, the 9.

2. Round to three significant figures.

(a) $\sqrt{13} = 3.6055512 \dots \approx \mathbf{3.61}$

(b) $8848.86 \text{ m} \approx \mathbf{8850 \text{ m}}$, better written $\mathbf{8.85 \times 10^3 \text{ m}}$

The third figure is the 4 in the tens place; the 8 after it rounds it up.

3. Scientific notation.

(a) $9,460,730,000,000 \text{ km} = 9.46073 \times 10^{12} \text{ km} \approx \mathbf{9.46 \times 10^{12} \text{ km}}$

(b) $0.000\,000\,000\,000\,000\,000\,160\,2 \text{ C} = 1.602 \times 10^{-19} \text{ C} \approx \mathbf{1.60 \times 10^{-19} \text{ C}}$

Six groups of three zeros is 18 zeros, so the 1 is in the 19th decimal place: the exponent is -19 . Write 1.60, not 1.6 — the trailing zero is the third significant figure.

4. Ordinary decimal numbers.

(a) $7.0 \times 10^{-5} \text{ m} = \mathbf{0.000\,070 \text{ m}}$

Keep the trailing zero: 7.0 has two significant figures and so must the decimal form. 0.00007 m loses one.

(b) $5.46 \times 10^7 \text{ km} = \mathbf{54,600,000 \text{ km}}$

5. Reading the ruler. The bar ends between the 8.3 and 8.4 cm marks, a little past halfway.

(a) **8.36 cm** — the 8.3 is certain and the final 6 is the estimated digit. *Accept 8.35 to 8.37.*

(b) **Three significant figures.**

(c) $8.36 \text{ cm} \times \frac{10 \text{ mm}}{1 \text{ cm}} = \mathbf{83.6 \text{ mm}}$

6. Convert.

(a) $508 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 1666.24 \text{ ft} \approx \mathbf{1670 \text{ ft}}$, or $\mathbf{1.67 \times 10^3 \text{ ft}}$

The published height is 1667 ft, so this checks.

$$(b) \ 310 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 86.111 \dots \approx \mathbf{86.1 \text{ m/s}}$$

7. Cheetah.

$$29 \frac{\text{m}}{\text{s}} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 64.854 \dots \approx \mathbf{65 \text{ mi/h}}$$

Two significant figures, set by the 29 m/s. Accept 64.9 mi/h for full credit on the conversion; the sig-fig point is worth discussing when the quiz is returned.

Early finisher

8. Radio command to Mars.

$$t = \frac{d}{v}: \quad 5.46 \times 10^7 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ s}}{3.00 \times 10^8 \text{ m}} \times \frac{1 \text{ min}}{60 \text{ s}} = 3.0333 \dots \approx \mathbf{3.03 \text{ min}}$$

That is 182 s each way at closest approach, and up to about 24 minutes when Mars is far — which is why a Mars rover cannot be driven with a joystick.

9. Widening of the Atlantic, 1492 to 2026: $2026 - 1492 = 534$ years.

$$534 \text{ yr} \times \frac{2.5 \text{ cm}}{1 \text{ yr}} \times \frac{1 \text{ m}}{100 \text{ cm}} = 13.35 \text{ m} \approx \mathbf{13 \text{ m}}$$

Two significant figures, set by “about 2.5 cm”; the 534 years is a count and does not limit the answer. About the length of a city bus — in five centuries.

10. Shower.

$$8.0 \text{ min} \times \frac{2.5 \text{ gal}}{1 \text{ min}} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 75.7 \text{ L} \approx \mathbf{76 \text{ L}}$$

Two significant figures: both 2.5 gal/min and 8.0 min have two.

11. Water per person per day.

$$\frac{1.0 \times 10^9 \text{ gal}}{8.5 \times 10^6 \text{ people}} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 445.29 \dots \text{ L} \approx \mathbf{450 \text{ L}}, \text{ better written } \mathbf{4.5 \times 10^2 \text{ L}} \text{ per person per day}$$

Both data values have two significant figures (“about 1.0 billion”, “about 8.5 million”), so the answer gets two. 445 L or 445.3 L overstates the precision; give full credit for the calculation and the sig-fig mark only for two figures.